



CANDIDATE NAME

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CT GROUP

20S7 ____

CENTRE NUMBER

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INDEX
NUMBER

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BIOLOGY

9816/01

Paper 1 Long Structured and Free-response Questions

Candidates answer on the Question Paper.

20 Sept 2021

Additional Materials: Insert

2 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the space provided on the Question Paper.

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 25
2	/ 25
3 / 4	/ 25
Final Mark	/ 75

This document consists of **16** printed pages and **5** blank pages.

BOOKLET I

Section A

Answer **all** the questions in this section.

QUESTION 1

You should read through the whole of the insert and Question 1 carefully and then use the information provided to answer the questions. The insert includes Fig. 1.1, Fig. 1.2, Fig. 1.3, Fig. 1.4, Fig. 1.5, Fig. 1.6 and Table 1.1.

- (a) The reward system in humans involves the neurotransmitter dopamine pathway.

Describe how dopamine acts as a neurotransmitter.

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[3]

- (b)(i) Outline **one** way that a signal can be transmitted from the brain to the other body parts.

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[2]

(ii) Explain how homeostasis is achieved as a result of leptin and ghrelin working together.

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[3]

(c)(i) Using the results of the study shown in Fig. 1.4, deduce the effect of FTO mutation on ghrelin concentration in the blood **and** explain how this affects the individual with the mutation.

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[4]

(ii) Comment on the validity of the results of this experiment.

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[2]

- (iii) Explain how the FTO enzyme is able to change the levels of ghrelin, a peptide hormone, on a molecular level.

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[3]

- (d) Based on the information provided in Fig 1.6 and Table 1.1, justify if the substrate-binding ability is abolished in the FTO protein coded for by the A allele.

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[4]

- (e)(i) Describe the role of the microbes that are normally found in humans.

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[2]

(ii) Suggest and explain a suitable gene to be cloned into the bacteria to treat obesity.

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..... [2]

[Total: 25]

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Candidate Name: _____

CT Group: 20S7 _____

BOOKLET II

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QUESTION 2

In a book published in 1905, HG Wells cited that 'biologically, the species is the accumulation of the experiments of all its successful individuals since the beginning'.

Discuss this statement.

[25]

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Candidate Name: _____

CT Group: 20S7 _____

BOOKLET III

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--- END OF SECTION B ---

--- END OF PAPER---

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